

Multiple Choice (10 marks)

1. As of 2020, which architecture is best for classifying high-resolution images?
 - (a) convolutional networks
 - (b) graph networks
 - (c) fully connected networks
 - (d) RBF networks
2. Which of the following tasks can be best solved using Clustering.
 - (a) Predicting the amount of rainfall based on various cues
 - (b) Detecting fraudulent credit card transactions
 - (c) Training a robot to solve a maze
 - (d) All of the above
3. Which one of the following is equal to $P(A, B, C)$ given Boolean random variables A, B and C, and no independence or conditional independence assumptions between any of them?
 - (a) $P(A | B) * P(B | C) * P(C | A)$
 - (b) $P(C | A, B) * P(A) * P(B)$
 - (c) $P(A, B | C) * P(C)$
 - (d) $P(A | B, C) * P(B | A, C) * P(C | A, B)$
4. A and B are two events. If $P(A, B)$ decreases while $P(A)$ increases, which of the following is true?
 - (a) $P(A|B)$ decreases
 - (b) $P(B|A)$ decreases
 - (c) $P(B)$ decreases
 - (d) All of above
5. Which of the following are the spatial clustering algorithms?
 - (a) Partitioning based clustering
 - (b) K-means clustering
 - (c) Grid based clustering
 - (d) All of the above

True / False (10 marks)

Answer True or False

6. True or False: K-fold cross-validation has a computational complexity that is quadratic in K, leading to significantly higher computation times as K increases.
7. True or False: To achieve the same projection as SVD in PCA, the data must be transformed to have a zero mean.
8. True or False: Out-of-distribution detection is commonly referred to as anomaly detection in machine learning.
9. True or False: The dimensionality of the null space of the matrix A is 2.
10. True or False: Applying an L_1 norm or L_0 norm penalty in linear regression can result in some coefficients being zeroed out.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function to remove characters from the first string which are present in the second string.
12. Write a function to divide two lists using map and lambda function.

End of Paper